



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to AADHINI AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

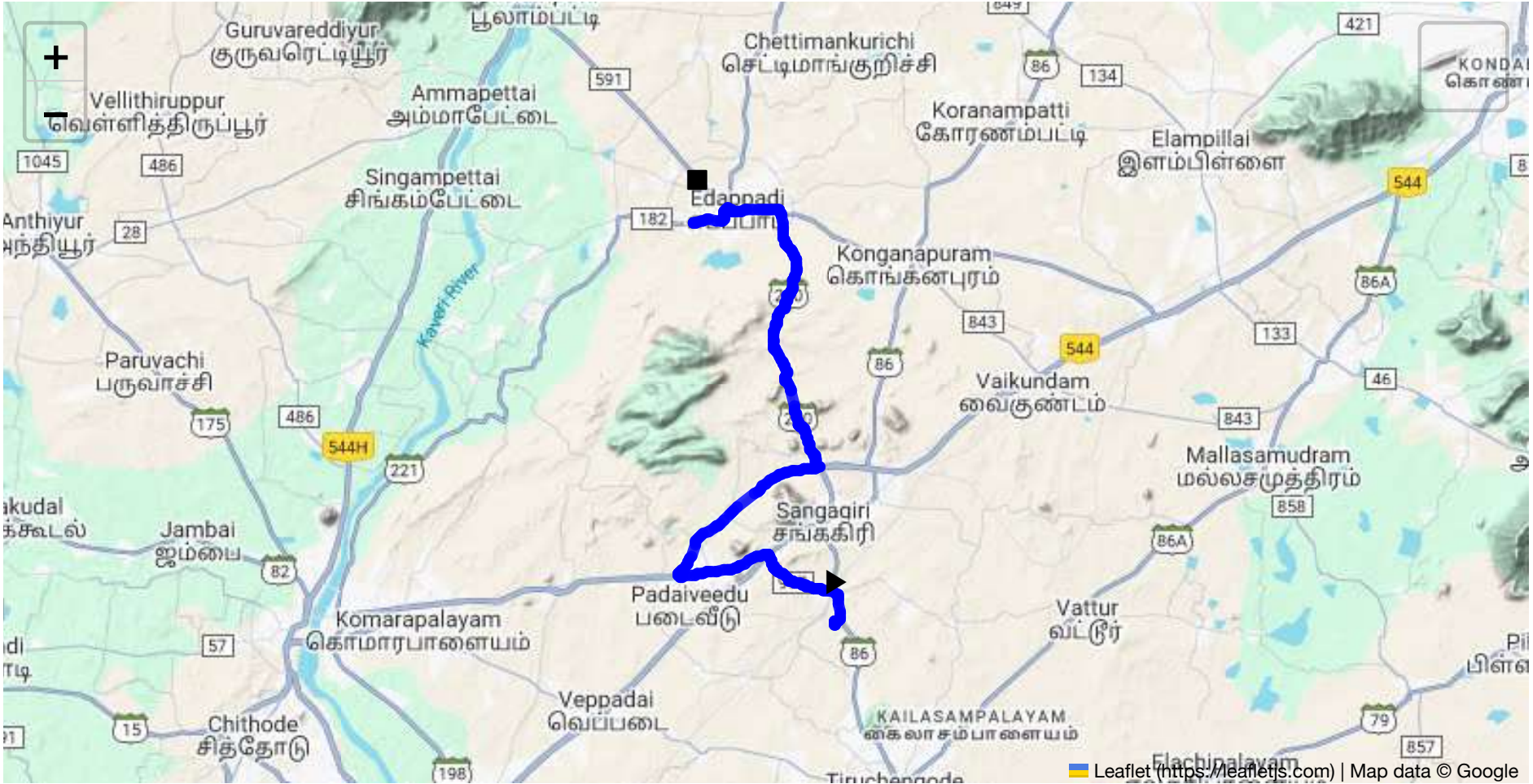
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 30.79 km
Estimated Duration: 0.7 hours
Adjusted Duration (Heavy Vehicle): 0.9 hours

Start: (11.4381, 77.8734)
End: (11.580409, 77.823288)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

The route from Sangagiri to Alachampalayam through Padaiveedu follows a path along the Bhavani Main Road in Tamil Nadu. It covers approximately 30.79 kilometers, starting from Sangagiri and passing through areas primarily characterized by rural landscapes, small towns, and agricultural regions.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

Tamil Nadu experiences a tropical climate with seasonal variations. The monsoon season, particularly from October to December, can bring heavy rains, leading to slippery roads and potential flooding, especially in low-lying areas. Summers, from March to June, can be extremely hot, which might affect vehicle performance if not properly managed.

3. Analysis of Traffic Patterns:

Traffic is typically moderate, with potential congestion points near town centers and marketplaces. Peak hours during the morning (8-10 AM) and evening (5-7 PM) can see increased traffic, especially near Sangagiri and any intermediate towns. Market days or local festivals can also significantly affect traffic flow.

4. Assessment of Road Quality and Infrastructure:

Road quality is generally good but may vary, with some sections potentially having potholes or uneven surfaces, especially following the monsoon. The main road infrastructure supports heavy vehicles, but caution is advised on narrower rural roads.

5. Suggestions for Alternative Routes for Emergencies:

In case of roadblocks or emergencies, consider using local internal roads that parallel the main route. Route plans should include checking for updates on local alternative roads in the area.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

Transportation of hazardous materials in Tamil Nadu is subject to strict regulations, including vehicle compliance checks and adherence to designated routes. Night travel restrictions may apply in certain areas, and it is essential to carry documented approvals for transporting hazardous materials.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

While specific historical incident data for this route may not be readily available, Tamil Nadu's roads occasionally experience accidents involving heavy vehicles due to overloading, driver fatigue, or poor road conditions, especially during adverse weather.

8. Environmental Considerations and Sensitive Areas:

The route passes through regions that may have ecologically sensitive areas, especially near water bodies. Drivers should minimize environmental impact by avoiding spillage and adhering to speed limits to reduce noise and emissions.

9. Analysis of Communication Coverage:

Telecommunication coverage is generally good throughout the route, but there may be occasional dead zones in rural stretches. Drivers should be equipped with reliable communication devices and emergency contact information.

10. Estimated Emergency Response Times for Different Route Segments:

Emergency response times can vary depending on proximity to urban centers. In towns like Sangagiri and Padaiveedu, response times may be quicker (approximately 20-30 minutes), while rural sections may experience delays (up to 45-60 minutes).

12. An Overall Summary of Risk Assessment:

The route from Sangagiri to Alachampalayam via Bhavani Main Road is generally manageable for heavy vehicles carrying hazardous materials when standard precautions are observed. Key risks include potential weather-related impacts, congested areas, and occasional road quality issues. Trips should be planned with contingency strategies, especially during adverse weather conditions or peak traffic times. Ensuring compliance with local regulations and preparedness for potential communication gaps is critical for safe transit.

Overall, strategic planning and situational awareness by truck drivers will mitigate most risks associated with this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58435, 77.83651	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.45353, 77.85697	30 KM/Hr
6	Turn	Medium	11.45839, 77.85143	30 KM/Hr
7	Turn	Medium	11.46307, 77.84941	30 KM/Hr
8	Turn	Medium	11.46318, 77.84913	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr
10	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
11	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
12	Turn	High	11.58533, 77.83450	15 KM/Hr
13	Turn	High	11.58052, 77.82229	15 KM/Hr
14	Blind Spot	Blind Spot	11.58039, 77.82224	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	KRS Hospital and Diabetic Centre	11.5848828, 77.8449664	30 km/h	Medium
2	hospital	Kannan Hospital	11.5854802, 77.8425936	30 km/h	Medium
3	hospital	Aravind Hospital	11.5854615, 77.8424165	30 km/h	Medium
4	hospital	Muthu Eyecare Hospital	11.584736, 77.8393794	30 km/h	Medium
5	clinic	Uma Clinic	11.58398, 77.83977	30 km/h	Medium
6	clinic	KGP Clinic	11.5847763, 77.8387488	30 km/h	Medium
7	hospital	Government Hospital, Edapadi	11.5848264, 77.8380174	30 km/h	Medium
8	hospital	Sri Sankar Hospital	11.5862984, 77.8355021	30 km/h	Medium
9	hospital	Sri Advika Hospital	11.585049, 77.834422	30 km/h	Medium
10	clinic	Raj Physiotherapy Clinic	11.5857357, 77.8341218	30 km/h	Medium
11	hospital	Dr. O.S. Subramanium Hospital	11.584996, 77.834235	30 km/h	Medium
12	hospital	Aravinthu Eye Hospital	11.5866631, 77.8341465	30 km/h	Medium
13	clinic	Xcellent Homecare Clinic	11.584762, 77.8338085	30 km/h	Medium
14	hospital	Mohan Hospital	11.5841331, 77.8336277	30 km/h	Medium
15	hospital	Nivetha Hospital	11.5858509, 77.833451	30 km/h	Medium
16	hospital	Rathanasamy Hospital	11.58583, 77.833443	30 km/h	Medium
17	hospital	SKM Hospital	11.5841611, 77.8334397	30 km/h	Medium
18	clinic	Krishna Hospital and Clinic	11.5846542, 77.8333239	30 km/h	Medium
19	hospital	Arunagiri Hospital	11.5837095, 77.8332129	30 km/h	Medium
20	hospital	Pachiappa Hospital	11.58571, 77.833131	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
21	clinic	Rakshan Polyclinic	11.5809367, 77.8326223	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58435, 77.83651



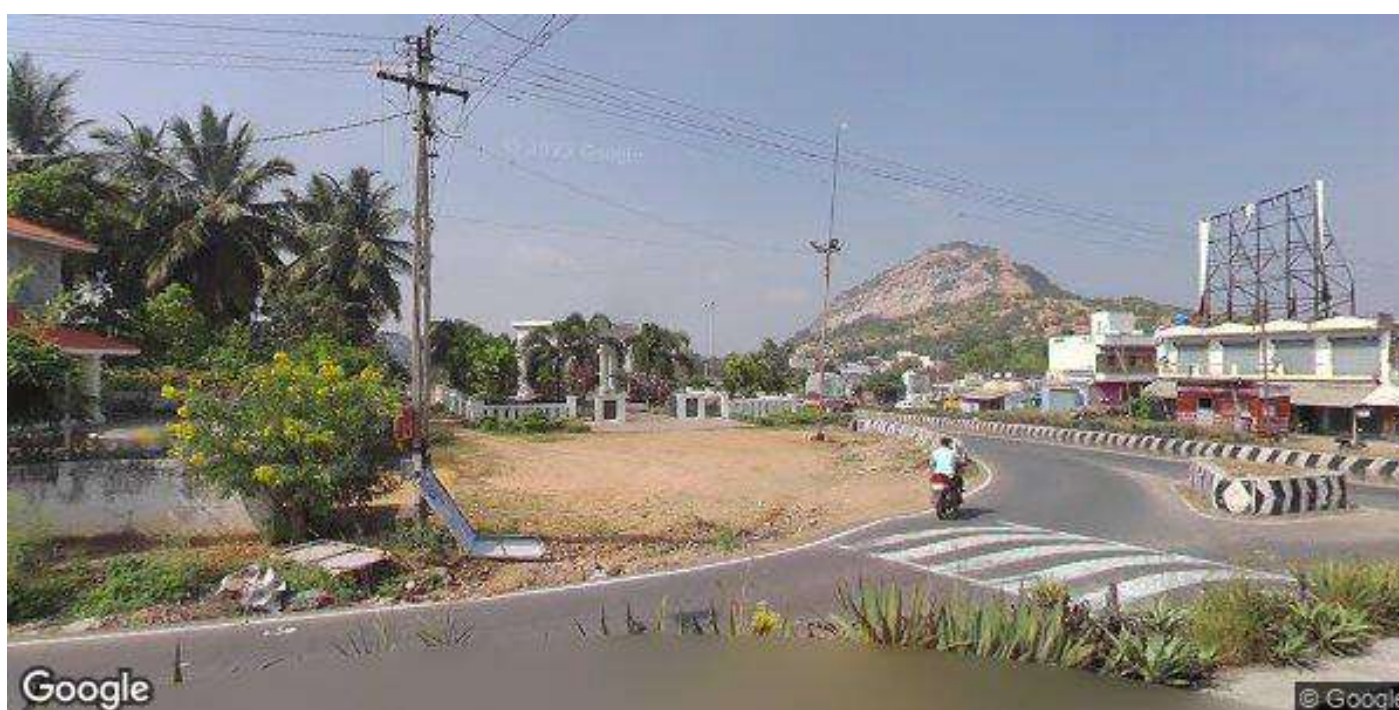
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46318, 77.84913



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49390, 77.86748



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58533, 77.83450



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58052, 77.82229



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.58039, 77.82224
